Sakdipat (Jaf) Kongkaeo

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Summary

Graduate student in Atmospheric and Oceanic Sciences at the University of Wisconsin–Madison focusing on El Niño-Southern Oscillation (ENSO) dynamics, seasonal predictability, and stochastic climate modeling using Linear Inverse Models (LIM/CSLIM).

Work Experience

Graduate Research Assistant Sep 2024 – present

- Conduct research on ENSO dynamics and predictability using Linear Inverse Modeling (LIM) and Cyclostationary LIM (CSLIM) frameworks.
- Develop MATLAB and Python scripts for EOF analysis, noise estimation, and seasonal climate experiments.
- Analyze global SST and SSH datasets to study mode structure, variance growth, and stochastic forcing.

Presentations

AGU Fall Meeting 2026 San Francisco, CA (Dec 7–11, 2026)

Poster Presentation: *"Disentangling the Role of Noise Seasonality in ENSO Variance using a Cyclostationary Linear Inverse Model."*

Projects

Seasonality of Noise Forcing and ENSO Predictability (LIM/CSLIM) Code

Analyzed ENSO dynamics and seasonal predictability using Linear Inverse Modeling (LIM) and Cyclostationary LIM (CSLIM). Computed seasonal noise covariance (Q), climatological variance (C_0), and their projections onto ENSO and non-ENSO mode adjoints. Implemented Floquet adjoint analysis, stochastic optimal structures, and 1-month forcing experiments to evaluate seasonal growth and variance amplification.

Education

2024 – present	Graduate student in Atmospheric and Oceanic Sciences, **University of Wisconsin–Madison**	
2021 – 2024	BS in Applied Physics, **University of Washington, Seattle**	(GPA: 3.92/4.00)
2019 – 2020	Grade 12, **Episcopal High School**, Alexandria, VA	
2014 – 2019	Secondary School, **Hatyaiwittayalai School**, Hat Yai, Thailand	

Teaching Experience

Teaching Assistant — ATM OCN 401 Fall 2026 – present

Grading assignments.

Teaching Assistant — ATM OCN 311 Spring 2026

Dynamics of the Atmosphere and Ocean II. Assisted with labs, supported student learning, and graded

assignments.

Physics Lab TA 2022 – 2023

Led undergraduate physics lab sessions at UW Seattle, guided students through experiments, and graded lab reports.

Skills

Programming	MATLAB, Python, R, Java
Data Analysis	EOF/PC analysis, regression modeling, time-series methods, climate reanalysis (ERA5, ORAS5, ERSSTv5)
Modeling	Linear Inverse Modeling (LIM), Cyclostationary LIM (CSLIM), stochastic forcing experiments
Communication	Scientific writing, presentation, teaching assistance

Awards & Honors

2019–present	**DPST Royal Thai Government Scholarship**	Institute for the Promotion of Teaching Science and Technology (IPST), Ministry of Education, Thailand
2020–2024	**Dean's List** (2020–21, 2021–22, 2022–23, 2023–24)	University of Washington, Seattle